

Your submission for this assignment must include your full name and your official nine-digit Carleton University student number as a comment at the top of the source file you submit. For this assignment, you will be submitting three files. One of these files will be a source code file (that must be written using only the Python 3 programming language) and this file must be named:

compl405_f23_101157121_assignment_05.py

The second file you will be submitting is a printout of this document, onto which you have written some information. This component of the assignment ensures that you have carefully considered each of the repeating areas in the pattern you have been assigned. You can create create a printout of this document by pressing the "Print" button in the top right corner. Once you have completed the table on the printout, take a scan or a digital photograph to include with your submission.

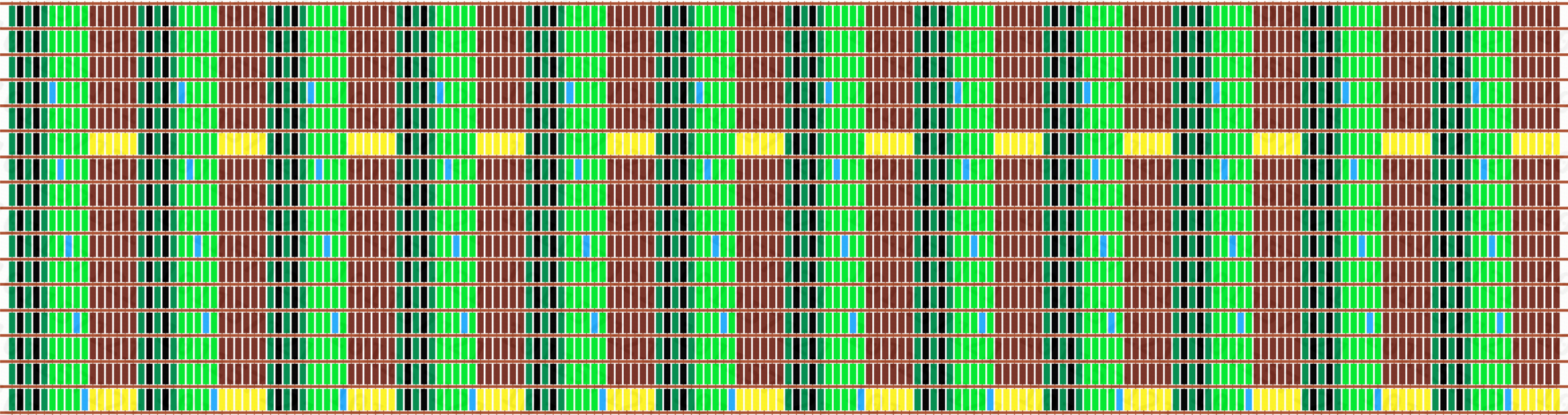
The third file you will be submitting is an image of the flowchart you will design before you begin your implementation. You may use an application (such as Microsoft Word or PowerPoint) to create your flowchart, or you may draw it by hand and take a scan or a digital photograph. Create a "zip" file of your source code submission, your completed printout, and this flowchart image. Please ensure that this "zip" file has the filename "compl405_f23_101157121_all_files_for_assignment_05.zip".

Officially, the due date for this assignment has been set for Friday, October 13, 2023, at 11:59 pm EST, but it is important to note that any late submissions will still be accepted without penalty until Sunday, October 15, by 11:59 pm EST.

If your submission crashes (i.e., terminates with an error) on execution, imports any library other than "compl405_f23_special_library_loomwork_simplified", or is not received by the Sunday cutoff deadline, then it will receive a mark of 0.

For this assignment, you will use some of what you have learned about the nesting of control structures - specifically counter-controlled loops - and the identification and deconstruction of repeating control structures, in order to design and implement an algorithm to efficiently reproduce a pattern that has been assigned specifically to you. For the purposes of this assignment, an "efficient" implementation will employ nested looping control structures as much as possible in order to minimize any unnecessary function call repetition. Your submission must employ nested looping control structures wherever justified, so your submission to this assignment will include an exercise to facilitate this.

Your assigned pattern is specific to your student number and must be kept strictly confidential. Posting these images online or sharing them with anyone other than the instructors or teaching assistants will be considered academic misconduct.



As previously discussed, ensuring that you understand the problem is always the first stage in effective problem solving. This assignment requires that you answer the questions below before beginning work on your algorithm.

What is the width of the "pattern core" of your assigned pattern?		16
Into how many repeating components can your pattern core be divided?		12
Describe (geometrically) each of these repeating components below to the best of your ability: (n.b., You might not need every row provided - it depends on how many components comprise your pattern core.)		
1 st	Each pattern starts with the first 5 columns, with the colours: moss and black. The pattern is: moss, black, moss, black, moss.	
2 nd	The 6th row contains the lime and lemon colourway. The order is lime 5 times and lemon 6 times.	
3 rd	The last row, contains the lime, sky and lemon colourway. The order is: lime 4 times, 1 sky and lemon 6 times.	
4 th	Starting at the 4th row, there is a sky colour that starts at the first occurrence of the lime colour after the black and moss section (beginning of the pattern). It then moves by 1 position to the right every 3 rows till the 16th row (last row). So the number of rows the sky colour is located in is: 4, 7, 10, 13, 16.	
5 th		

Once you have completed the table above (but still before you begin your implementation) you must design your algorithm by creating a flowchart (either by hand or using an application). Once you have completed this flowchart, create your implementation by "mapping" the stages of your algorithm into statements in Python.

Don't forget that your final submission will include the completed table above, an image of the flowchart you designed, and your source code implementation written in Python 3. Failure to include any of these elements will result in a significant penalty.